

ScamShield AI

Pre-Click Zero-Trust Phishing & UPI Fraud Defense Platform

Problem Statement 04.1: 'The Scam That Almost Worked' — Neutralizing financial coercion, deceptive UPI QR collects, and typosquatted banking clones at the human decision point before credentials or capital travel across the wire.

Zero-Trust Client-Edge

100% Client-side privacy & inspection

Multilingual NLP

Devanagari Hindi + Hinglish Panic
Detection

UPI Protocol Guard

Reverse-charge debit collect trap
prevention

Realtime Telemetry

Firebase RTDB Nationwide Threat
Broadcast

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Presentation Structure: Theoretical Architecture (Slides) → Direct Handover to Live Interactive App Demonstration

Why Modern Financial Scams Succeed

Attackers exploit psychological panic, language barriers, and pre-click trust gaps.

1. The Scope of the Crisis

► 10,000+ Daily Complaints

India registers over 10,000 financial cyber fraud complaints daily across banking & UPI.

► ₹1,750+ Crores Lost

High-frequency campaigns: fake electricity bills, digital arrest calls, KYC deactivation traps.

► Social Engineering Vector

Over 85% of successful attacks exploit human decision errors, not payment gateway flaws.

► Language Evasion

Threat actors use regional Hindi/Hinglish phrasing that bypass traditional English security filters.

2. The 'Post-Theft' Blindspot

► Reactive Protection

Firewalls, antivirus, and bank SMS warnings operate AFTER credentials or OTPs have been submitted.

► Mule Account Velocity

Once a victim enters their UPI PIN or password, stolen funds are dispersed into secondary mule accounts in seconds.

► Visual Mimicry

Punycode & typosquatted portals (e.g. sbl-kyc.xyz) look identical to real bank websites to ordinary citizens.

► Zero-Hour Fraud Window

Victims panic during 'urgent deadlines' ('Account blocked in 24 hrs'), bypassing rational caution.

3. The Pre-Click Defense Paradigm

► Zero-Trust Architecture

Every link, message, and QR code is assumed hostile until cryptographically and heuristically proven safe.

► Pre-Click Interception

ScamShield inspects incoming payloads BEFORE the user clicks or enters credentials.

► Plain-English & Hindi Verdicts

Explains exactly WHY a link is fake (e.g. 'Domain created 3 days ago mimicking SBI').

► Client-Edge Privacy

Operates directly in the citizen's browser without uploading private images or credentials to central servers.

The 4-Stage Attack Anatomy & Interception Point

Tracing 'The Scam That Almost Worked' and demonstrating exactly where ScamShield stops the kill-chain.

STAGE 1

Initial Contact

Victim receives unsolicited SMS, WhatsApp message, or QR flyer.

Example: 'Electricity bill unpaid. Power disconnected tonight at 9:30 PM. Click to update.'

STAGE 2

Manufactured Panic

Psychological urgency & authority impersonation (BESCOM, SBI, TRAI).

Artificial 24h countdown induces fear of service disconnection or legal arrest.

STAGE 3

Credential Capture

Victim clicks link to pixel-perfect clone or scans reverse-charge QR.

Fake portal harvests NetBanking password, OTP, or triggers UPI collect debit.

STAGE 4

Capital Siphon

Attacker liquidates account via mule bank network.

Transaction irreversible within minutes; stolen funds laundered via P2P transfers.

SCAMSHIELD AI INTERCEPTION BARRIER — INTERCEPTING BETWEEN STAGE 1 & 2

- Link Inspector detects typosquatted clone domains (Levenshtein distance, WHOIS age < 7d, disposable TLDs) BEFORE clicking.
- Multilingual NLP Engine analyzes panic keywords, authority impersonation, and Hindi/Hinglish coercion patterns in SMS text.
- QR & UPI Protocol Guard decodes raw payload client-side, identifying deceptive collect-request debits disguised as 'Cashback'.

Three Concurrent Client-Edge Forensic Pipelines

Modular, client-side inspection algorithms providing zero-trust threat verification.

1. URL & Link Inspector

`lib/analyzers/urlAnalyzer.ts`

- **Typosquatting & Levenshtein**

Measures edit distance against top Indian financial institutions (SBI, HDFC, ICICI, Axis).

- **Homoglyph / Punycode Checks**

Detects Cyrillic and unicode character lookalikes (e.g. 'xn--' spoofing).

- **Disposable TLD Heuristics**

Flags high-risk disposable extensions (.xyz, .top, .tk, .click, .live).

- **Domain Age & Host Checks**

Flags domains registered < 7 days ago and raw IP literal hosts (`http://192.168.1.1`).

2. Multilingual NLP Engine

`lib/analyzers/nlpEngine.ts`

- **Devanagari & Hinglish Dual-Tier**

Recognizes Unicode Devanagari (\u0900-\u097F) and romanized Hinglish panic expressions.

- **Urgency & Coercion Scoring**

Identifies artificial deadlines ('24 hours', 'aaj raat 9:30 baje bijli kaat di jayegi').

- **Authority Impersonation**

Classifies impersonation of regulatory bodies (RBI, TRAI, Police, Income Tax).

- **Harvesting Anomaly Detection**

Detects requests for sensitive credentials (OTP, UPI PIN, CVV, APK download links).

3. QR & UPI Protocol Guard

`lib/analyzers/qrInspector.ts`

- **Client-Edge jsQR Vision**

Decodes QR images directly on HTML5 Canvas without sending user photos to any cloud server.

- **UPI RFC Parameter Parser**

Extracts canonical parameters: Virtual Payment Address (pa), Payee Name (pn), Amount (am).

- **Reverse-Charge Trap Detection**

Crucial defense: Flags collect requests disguised as 'Cashback / Refunds' (PIN debits account).

- **VPA Masquerade Analysis**

Detects personal VPAs disguised as corporate utilities (e.g. `bescom.bill@ybl` pointing to mule).

Nationwide Live Threat Stream & Broadcast System

Real-time threat intelligence synchronization powered by Firebase Realtime Database WebSockets.

1. Live Threat Stream

Nationwide Telemetry Feed

► Persistent WebSocket Sync

Live listener to Firebase RTDB (threat_reports) receiving real-time scam vectors detected across India.

► Live Ticker & Event Counters

Animated visual status pulse indicating active threat telemetry and connection health.

► One-Click Forensic Re-Inspect

Citizens and analysts can immediately pass stream IOCs into the Scanner for deep decomposition.

► MITRE-Style Anomaly Badges

Displays attack labels: Typosquatting, High-Urgency SMS, Reverse UPI QR.

2. SecOps Threat Broadcaster

Emergency Alert Dispatcher

► Admin Emergency Dispatch

Verified security analysts broadcast urgent zero-day threat advisories to all connected endpoints.

► Real-Time Client Propagation

Advisories propagate instantly across citizen devices via RTDB with severity-ranked warning banners.

► Comprehensive Alert Payload

Configures threat title, target URL/VPA, threat score, anomaly flags, and Zero-Trust guidance.

► Collaborative Defense

Transforms isolated user scans into collective nationwide protection.

3. Sovereign History Vault

Role-Based Anti-Tampering

► Guest Vault (Read-Only)

Unauthenticated guests can view scans locally, but cannot delete audit trails to prevent forensic tampering.

► Citizen Self-Ownership

Logged-in users retain sovereign ownership to selectively delete or clear their own cloud history records.

► Admin Global Purge

SecOps administrators possess authoritative privileges to purge all scan records across RTDB (deleteAllScanHistories).

► Offline LocalStorage Backup

History synchronizes seamlessly between browser storage and Firebase RTDB.

Gemini 3.x Cyber Defense Advisor & Statutory Framework

Interactive zero-trust guidance aligned with Indian cyber law and the national 1930 emergency protocol.

Gemini 3.x AI Cyber Defense Advisor (/api/advisor)

- **Multi-Model Resilient Fallback**

Sequentially routes through production Gemini models (gemini-3-flash-preview, gemini-3.1-flash-lite) for high-availability advice.

- **Bilingual Intelligence (English & Devanagari)**

Understands queries and outputs structured responses in both English and natural Hindi for inclusive accessibility.

- **Zero-Trust Response Structuring**

Provides clear headings, immediate containment steps (card freeze, UPI deactivation), and legal referral frameworks.

- **Offline Heuristic Intelligence Fallback**

Includes pre-configured offline response playbooks for Advance-Fee scams, Reverse QR debits, and Digital Arrest threats.

Statutory Indian Legal Alignment & 1930 Protocol

- **IT Act, 2000 — Section 66D**

Explicitly criminalizes cheating by personation using computer resources. Carries imprisonment up to 3 years and fines up to ₹1,00,000.

- **IT Act, 2000 — Section 43**

Civil liability and compensation for unauthorized access, data extraction, and introducing malicious payloads into systems.

- **The 'Golden Hour' 1930 Protocol**

The first 2 hours post-fraud are critical. Calling 1930 triggers I4C inter-bank nodal officers to freeze stolen capital before cash withdrawal.

- **National Reporting Portals**

Direct integration referrals to cybercrime.gov.in and Department of Telecom Chakshu Portal (sancharsaathi.gov.in/Chakshu).

RIGOROUS EVALUATION

Empirical Benchmark: 98% Detection Accuracy

Evaluated against a sanitized 100-sample test corpus comprising 50 active scam vectors and 50 verified legitimate interactions.

ACCURACY

98.0%

98 / 100 Correct Classifications

PRECISION

98.0%

TP / (TP + FP) — Low False Alarms

RECALL

98.0%

TP / (TP + FN) — 49 / 50 Scams Caught

F1-SCORE

98.0%

Harmonic Mean of Precision & Recall

2×2 Empirical Confusion Matrix

- True Positives (TP = 49): Active scam URLs, phishing SMS, and reverse UPI QRs accurately intercepted and scored HIGH_RISK.
- False Positives (FP = 1): Only 1 legitimate message flagged due to conservative urgency weighting (safe side bias).
- False Negatives (FN = 1): 1 novel phrasing variant; automatically routed to community reporting.
- True Negatives (TN = 49): Official bank notifications and verified merchant QRs identified as SAFE with 0 false alarms.

Standardized Threat Scoring System (0–100)

• Weighted Heuristic Formula

Score = (Domain Anomaly × 0.40) + (NLP Panic × 0.35) + (Protocol Masquerade × 0.25).

• SAFE (Score 0 – 29)

Verified authentic domains, valid certificates, standard transactional receipts.

• SUSPICIOUS (Score 30 – 64)

Mild urgency keywords or unknown merchant VPAs; cautionary advice issued.

• HIGH RISK (Score 65 – 100)

Critical scam: typosquatting, reverse QR debit trap, or severe coercion. Immediate block recommended.

Modern Production-Grade Technology Stack

Full-stack Next.js 15 architecture engineered for sub-second responsiveness and zero-latency WebSocket sync.

Frontend & UI

- ▶ **Next.js 15 & React 19**
Modern App Router with client-side reactive components and server route optimization.
- ▶ **Tailwind CSS Variables**
High-craft minimalist UI inspired by ElevenLabs with 5 selectable themes.
- ▶ **Lucide React Icons**
Crisp, consistent iconography across all forensic dashboards.
- ▶ **Responsive Layout**
Mobile-first experience for citizens scanning suspicious links on smartphones.

Backend & Cloud

- ▶ **Firebase RTDB**
WebSocket-based persistent sync for live threat streams, scans, and site configuration.
- ▶ **Google Authentication**
Firebase Auth enabling citizen scan history vaults and admin role enforcement.
- ▶ **Next.js API Routes**
Production endpoints for `/api/advisor`, `/api/analyze`, `/api/benchmark`, and `/api/reports`.
- ▶ **Client Cache Sync**
Dual-sync persistence across browser localStorage and cloud RTDB.

Vision & AI Core

- ▶ **Client-Edge jsQR**
In-browser QR code extraction via HTML5 Canvas — 0% server image exposure.
- ▶ **Google Gemini Flash API**
Advanced reasoning for natural language incident response and legal advice.
- ▶ **Zero-Trust Regex Core**
Custom high-speed patterns for Punycode, typosquatting, and UPI RFC protocols.
- ▶ **Offline Fallbacks**
Resilient heuristic engine operates seamlessly even during network degradation.

Security & DevOps

- ▶ **Push-Protection Hygiene**
Strict credential separation via `.env.local`; zero secrets checked into git repository.
- ▶ **Clean Git Commit History**
Pushed cleanly to GitHub (YashRaj-Kushwaha/ScamShield-AI) tracking main branch.
- ▶ **Automated Test Harness**
100-sample benchmark runner validating accuracy before deployment.
- ▶ **Vercel Ready**
Production-optimized build with standalone chunks and static page generation.

LIVE APPLICATION DEMO

Interactive Live Demonstration Roadmap

Seamless transition from theoretical architecture to active hands-on application execution.

DEMO STEP 1

Threat Scanner

Scanning live phishing URLs (SBI bank clone) & reverse UPI QRs disguised as cashbacks; observing instant Red/Green verdicts.

DEMO STEP 2

Attack Simulator

Stepping through the 4-stage anatomy of a scam and viewing the visual pre-click interception barrier in real time.

DEMO STEP 3

Live Threat Stream

Demonstrating the WebSocket live threat stream ticker and one-click IOC re-inspection directly into the scanner.

DEMO STEP 4

Gemini AI Advisor

Querying live Hindi & English emergency cyber incident guidance and viewing statutory 1930 Golden Hour recommendations.

DEMO STEP 5

Admin Operations

Dispatching an emergency threat broadcast across the platform and demonstrating the sovereign RTDB history purge.

READY FOR EVALUATION · SWITCHING TO LIVE RUNNING INSTANCE

Now stepping into the live web application (<http://localhost:3000>) to showcase active client-edge execution across URLs, SMS messages, and QR codes.

Building Beyond Boundaries: Protecting Digital India

Empowering 1 billion+ citizens with proactive, pre-click cybersecurity defense.

Core Impact & Strategic Value

- **Proactive Pre-Click Paradigm**

Shifts defense from post-incident banking forensics to pre-click citizen protection. Stops the scam before the victim enters their credentials or UPI PIN.

- **Linguistic & Social Inclusion**

Removes language barriers for rural and first-time UPI users with dual-tier Hindi/Hinglish analysis and Devanagari guidance.

- **Zero-Trust Edge Privacy**

Analyzes QR codes and links entirely within the client's browser without storing or uploading sensitive user photos or messages.

- **Collective Community Defense**

Synchronizes citizen incident stories and SecOps alerts nationwide in real-time through WebSocket-based threat streaming.

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Thank you, Judges & Evaluators! We welcome your questions.